

DOGUPARTHI AAKASH

Aspiring AI &ML Engineer

Portfolio:

portfolio-two-phi-3g2fmomh8y.vercel.app

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PROFESSIONAL SUMMARY

Motivated final-year Computer Science undergraduate (AI & ML) passionate about low-level systems, OS development, and applied AI/ML. Creator of the Mithl OS and UOS mobile OS built from scratch, and inventor of the NUX programming language — demonstrating deep interest in systems architecture and performance engineering. Proficient in Python, Java, and modern web technologies, with hands-on experience across full-stack development, machine learning pipelines, and compiler/runtime design.

EDUCATION

B.E—Computer Science & Engineering (AI & ML)

V.S.B. Engineering College, Karur

CGPA: 7.46 / 10.0

Class XII: 129.5 cutoff

Govt. HRSC School, Periya Kulathu Palayam, Karur

Class X

Venus Matriculation School, Karur

TECHNICAL SKILLS

Languages: Python, Java, C, Rust, Assembly (NUX lang development)

Frontend: HTML5, CSS3, JavaScript, React, TypeScript, Tailwind CSS

Backend & Databases: Flask (Python), Java, SQLite, SQL, Supabase

AI / ML: YOLO (object detection), Spam Detection, TF-IDF, Classification models

Systems & OS: OS development (Mithl OS, UOS/FreeBSD), Virtual Machine design (Plaza), Compiler/runtime design (NUX)

Tools & Platforms: Git, GitHub, VS Code, PyCharm, Eclipse, Google Colab, Canva, Adobe Express, Microsoft Office

PROJECTS

Clerk <https://github.com/DoguparthiAakash/clerk>

- Building a multi-agent AI system that combines code generation, repository analysis, debugging, documentation, DevOps automation, and project management into a unified intelligent software engineering platform. Designed with scalable agent orchestration, semantic code search, long-term memory, local and cloud LLM support, and IDE integration to automate complex engineering workflows while maintaining developer control.

UOS— Mobile OS <https://github.com/DoguparthiAakash/freebsd-src/tree/mobile-os/mobile>

- Alternative mobile operating system based on FreeBSD, delivering an iOS-like experience for mobile devices.
- Designed for high stability and performance with comprehensive architecture support across ARM (MediaTek, Qualcomm) and RISC-V processors.

NUX Programming Language https://github.com/DoguparthiAakash/Nux_Lang

- A programming language inspired by C, Rust, and Assembly — designed for simplicity, high execution speed, and memory safety.
- Bridges the gap between low-level control and modern language ergonomics.

Plaza VM <https://github.com/DoguparthiAakash/plazavm>

- Building a Rust-based virtualization platform that unifies virtual machines, Linux containers, and isolated development environments into a single intelligent workspace inspired by modern container and orchestration technologies.
- Features autonomous resource management, workload scheduling, snapshotting, live environment provisioning, cross-platform architecture support, and AI-assisted optimization for scalable software development and testing.

INTERNSHIPS

Developer — Emglitz Technologies

- Built a Spam Detection system using supervised machine learning techniques to classify emails as legitimate or spam.
- Handled data preprocessing (tokenization, TF-IDF vectorization) and trained classification models achieving reliable detection accuracy.

Project Manager — Infosys Springboard

- Led development of a Smart Home Energy Manager, overseeing requirements analysis, iterative delivery, and team coordination.
- Gained practical exposure to project lifecycle management and professional software delivery workflows.

Administrator — ServiceNow Xanadu

- Supported ServiceNow administration by managing users, roles, incidents, and service requests while ensuring platform stability. Gained hands-on experience in workflow configuration, ITSM processes, and system troubleshooting.

CERTIFICATIONS

- Artificial Intelligence 2.0 — GUVI
- Java & Python Certification — Infosys Springboard
- Python, SQL, HTML/CSS/JS, Soft Skills — Bitlabs
- Fundamentals of Machine learning and deeplearning — NPTEL

EXTRA-CURRICULAR & ACHIEVEMENTS

- Designed creative visual content (posters, presentations, event materials) using Canva and Adobe Express for college events.
- Actively solving algorithmic problems on LeetCode to strengthen data structures and competitive programming skills.
- Certified fast typist at 42 WPM — enables efficient documentation, data entry, and productive coding workflows.
- Proficient in Microsoft Office Suite (Word, Excel, PowerPoint) for documentation, data reporting, and presentations.